



Welcome to the Guition Workshop!

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Getting Started

Introduction

The objective of this post is to explain how to upload an Arduino program to the JC3636W718 module, from guition .

For secondary development, it is necessary to turn off HID in the settings, then restart the device or press and hold BOOT to enter download mode.

<http://www.guition.com/>

ESP32-S3-N16R8 has a significant performance improvement. It is equipped with a high-performance dual-core Tensilica LX7 MCU. One core handles high speed connection and the other for standalone application development. The dual-core MCU has a 240 MHz frequency and a computing power of 600 DMIPS.

Installing using Arduino IDE

Programming the ESP32

An easy way to get started is by using the familiar Arduino IDE. While this is not necessarily the best environment for working with the ESP32, it has the advantage of being a familiar application, so the learning curve is flattened.

We will be using the Arduino IDE for our experiments.

1, Installing using Arduino IDE

we first need to install version 1.8.19 of the Arduino IDE (or greater),for example, the Arduino installation was in "C:/Programs(x86)/Arduino".

download release link:

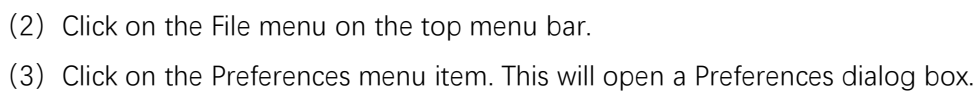
<https://downloads.arduino.cc/arduino-1.8.19-windows.exe>

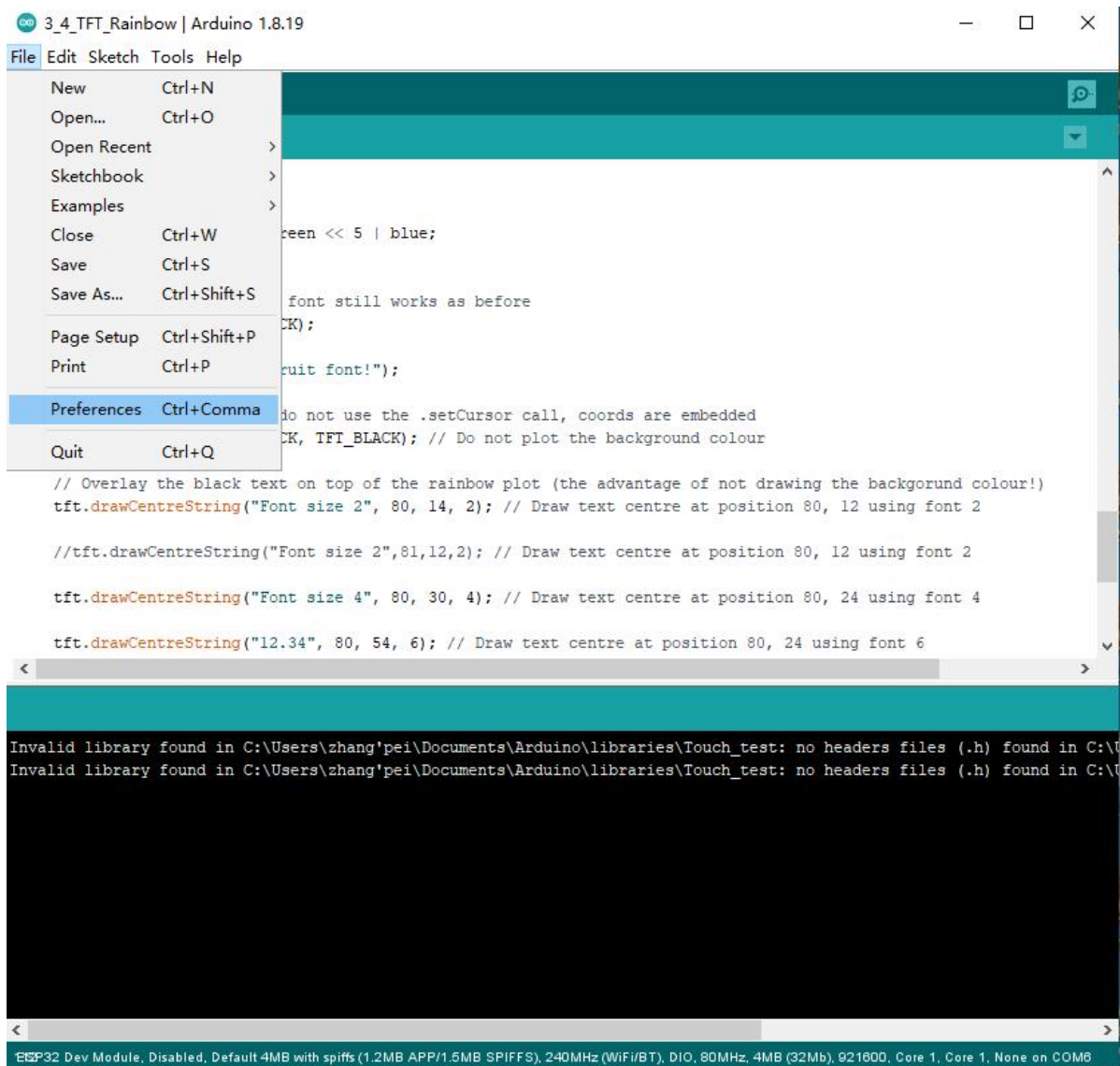
2, This is the way to install Arduino-ESP32 directly from the Arduino IDE.

Add Boards Manager Entry

Here is what you need to do to install the ESP32 boards into the Arduino IDE:

- (1) Open the Arduino IDE.





(4) You should be on the Settings tab in the Preferences dialog box by default.

(5) Look for the textbox labeled "Additional Boards Manager URLs".

(6) If there is already text in this box add a comma at the end of it, then follow the next step.

(7) Paste the following link into the text box :

Stable release link:

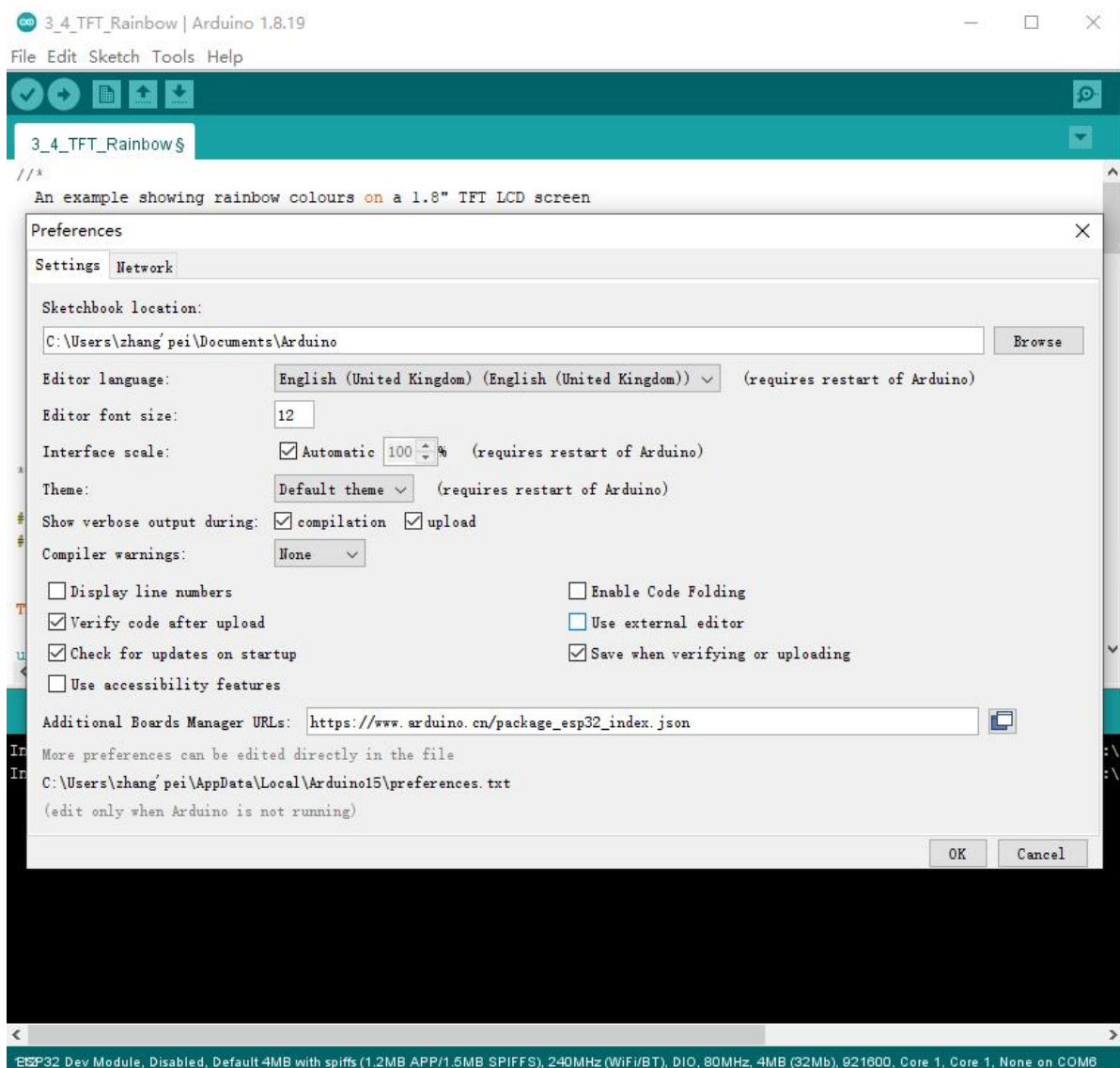
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

Development release link:

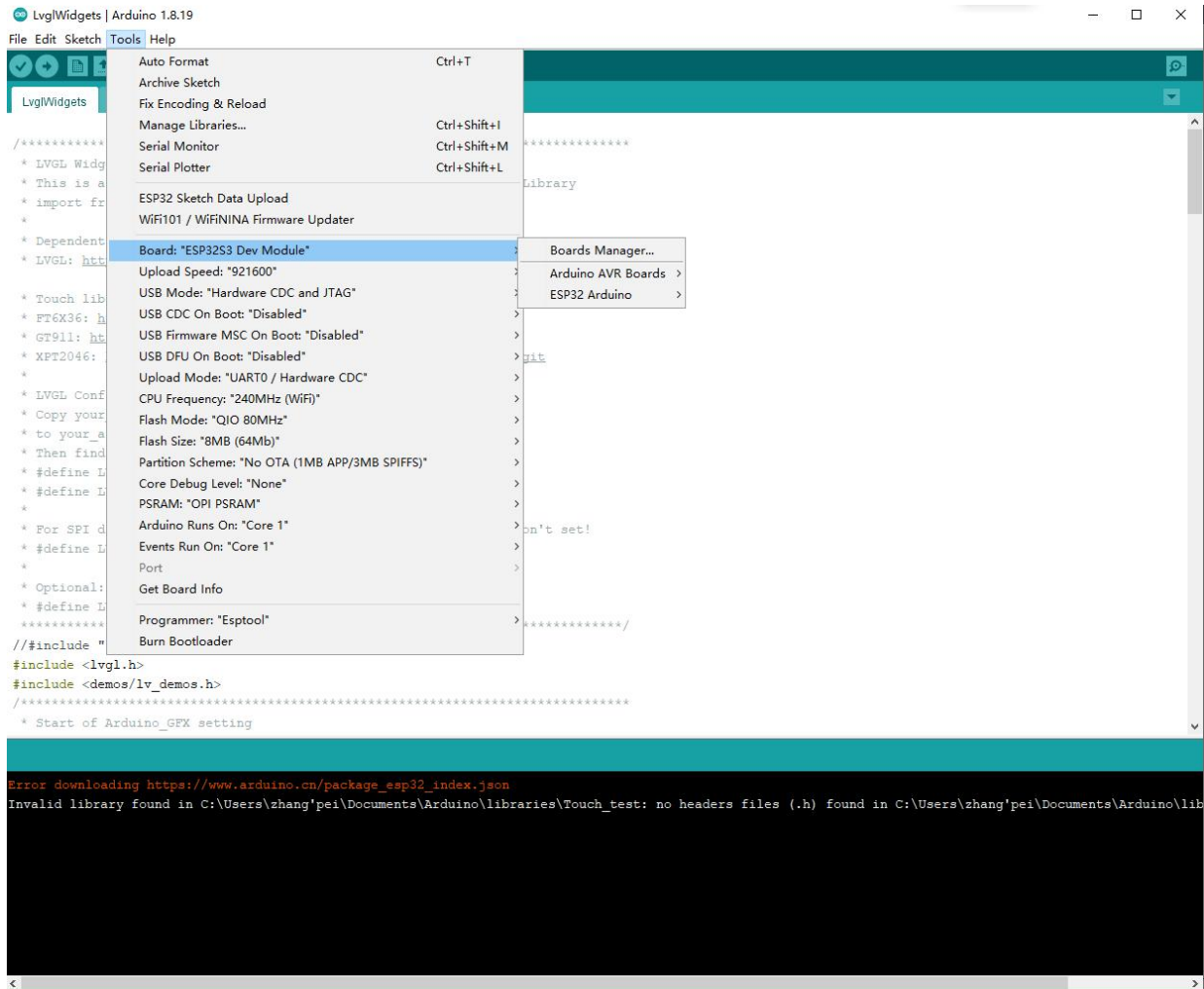
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_dev_index.json

(8) Click the OK button to save the setting.

The textbox with the JSON link in it is illustrated here:

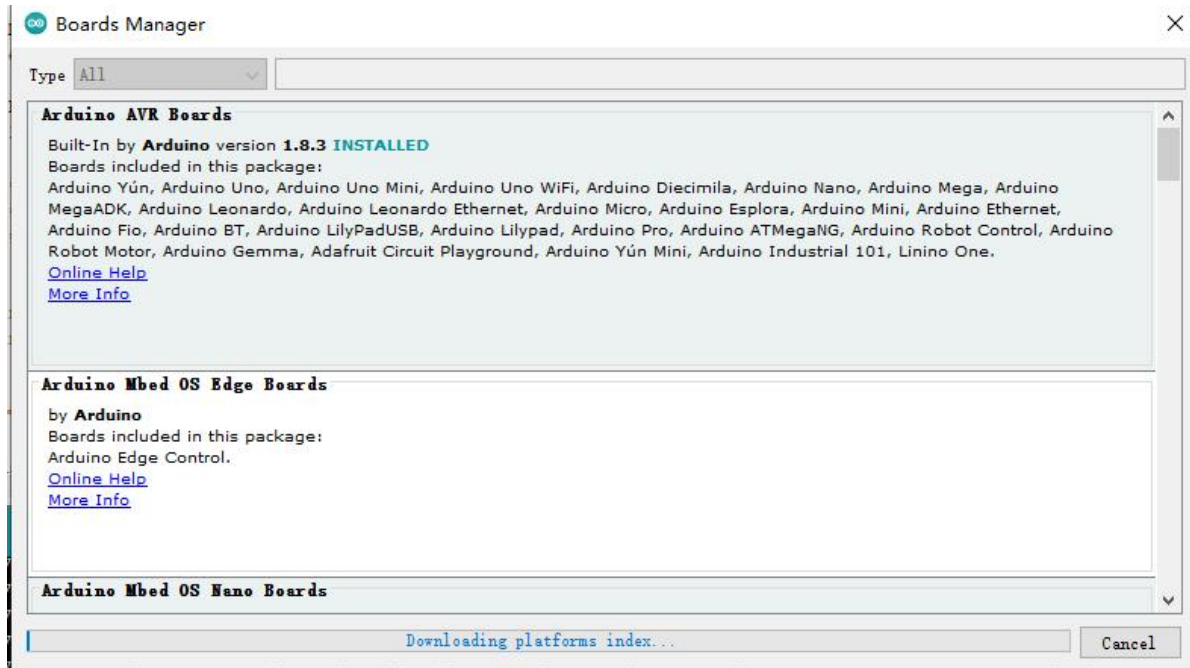


- (9) In the Arduino IDE click on the Tools menu on the top menu bar.
- (10) Scroll down to the Board: entry
- (11) A submenu will open when you highlight the Board: entry.
- (12) At the top of the submenu is Boards Manager. Click on it to open the Boards Manager dialog box.
- (13) In the search box in the Boards Manager enter "esp32".

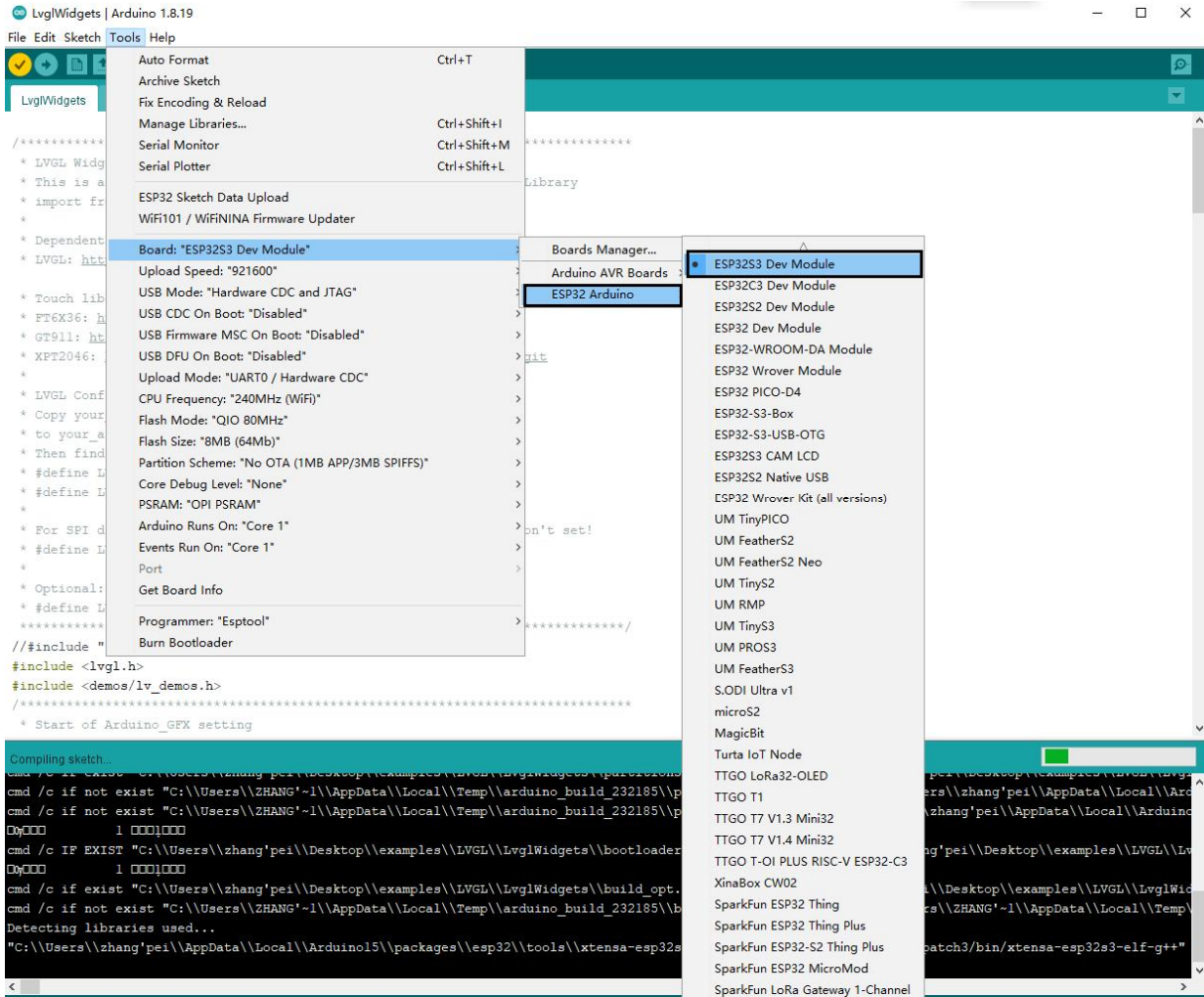


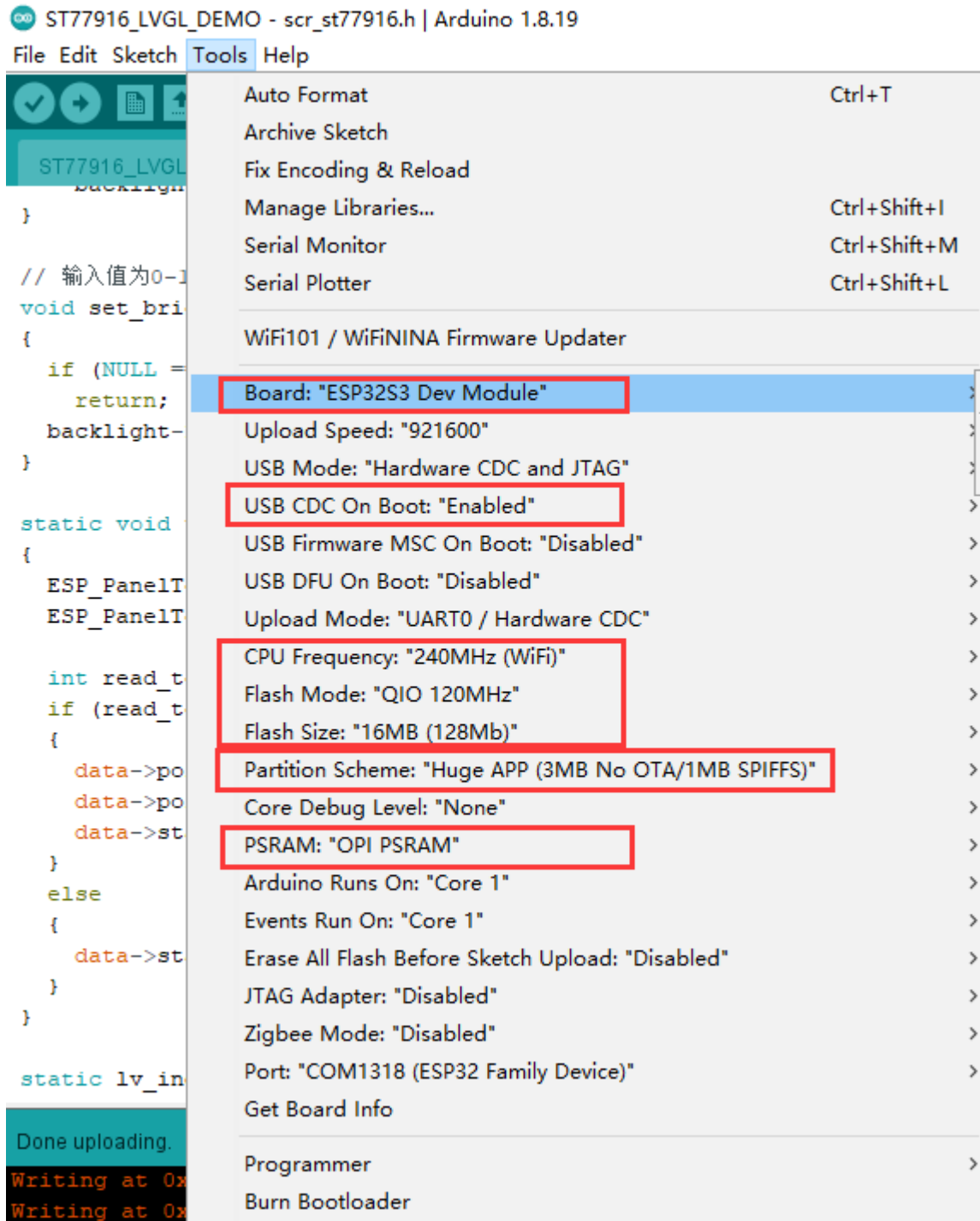
(14) You should see an entry for “esp32 by Espressif Systems”. Highlight this entry and click on the Install button.

This will install the ESP32 boards into your Arduino IDE

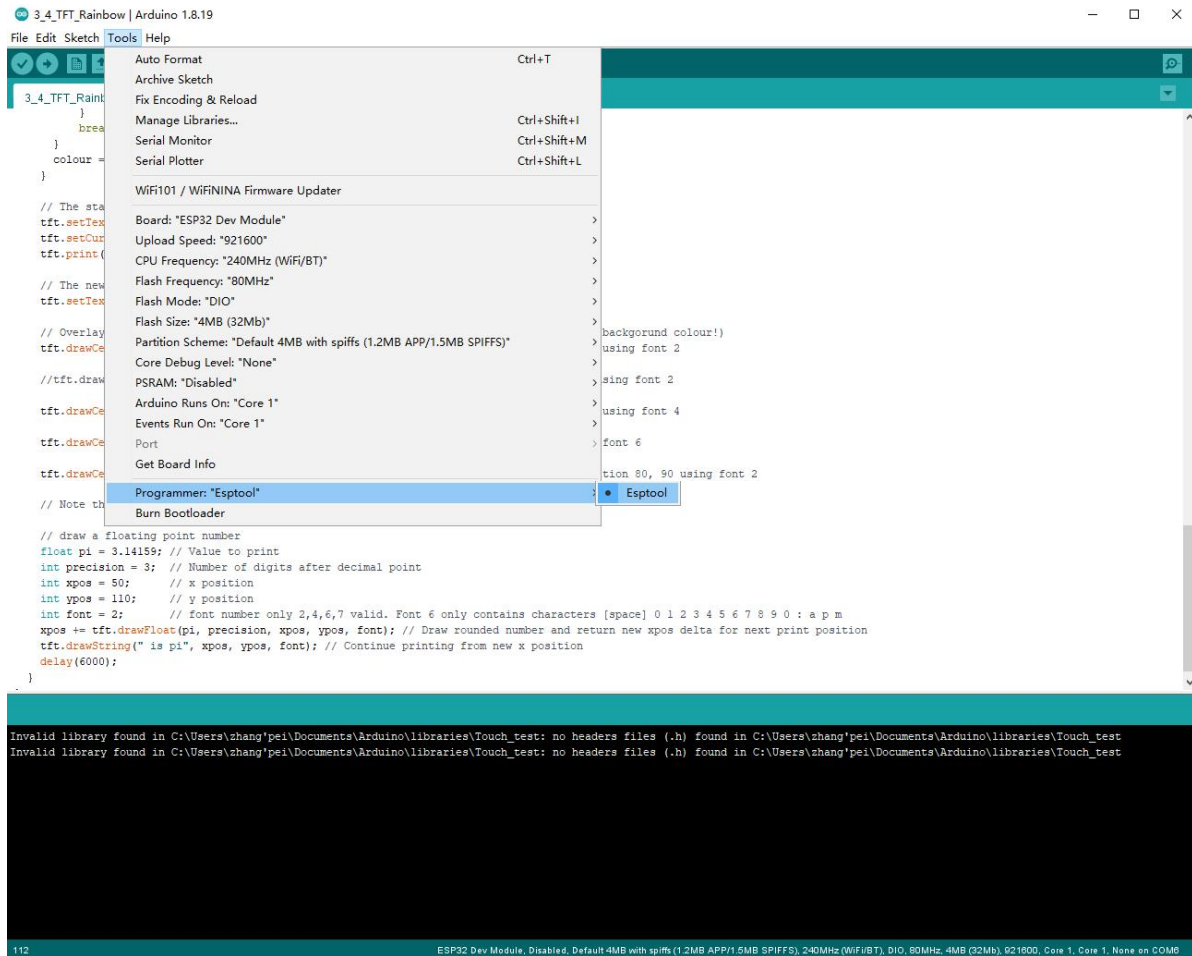


Once the installation completes, we need to select the correct board options for the "ESP32 Arduino" board. In the board type, in the tools tab, we choose "ESP32S3 Dev Module".





Set and In the programmer entry of the same tab, we choose "esptool".



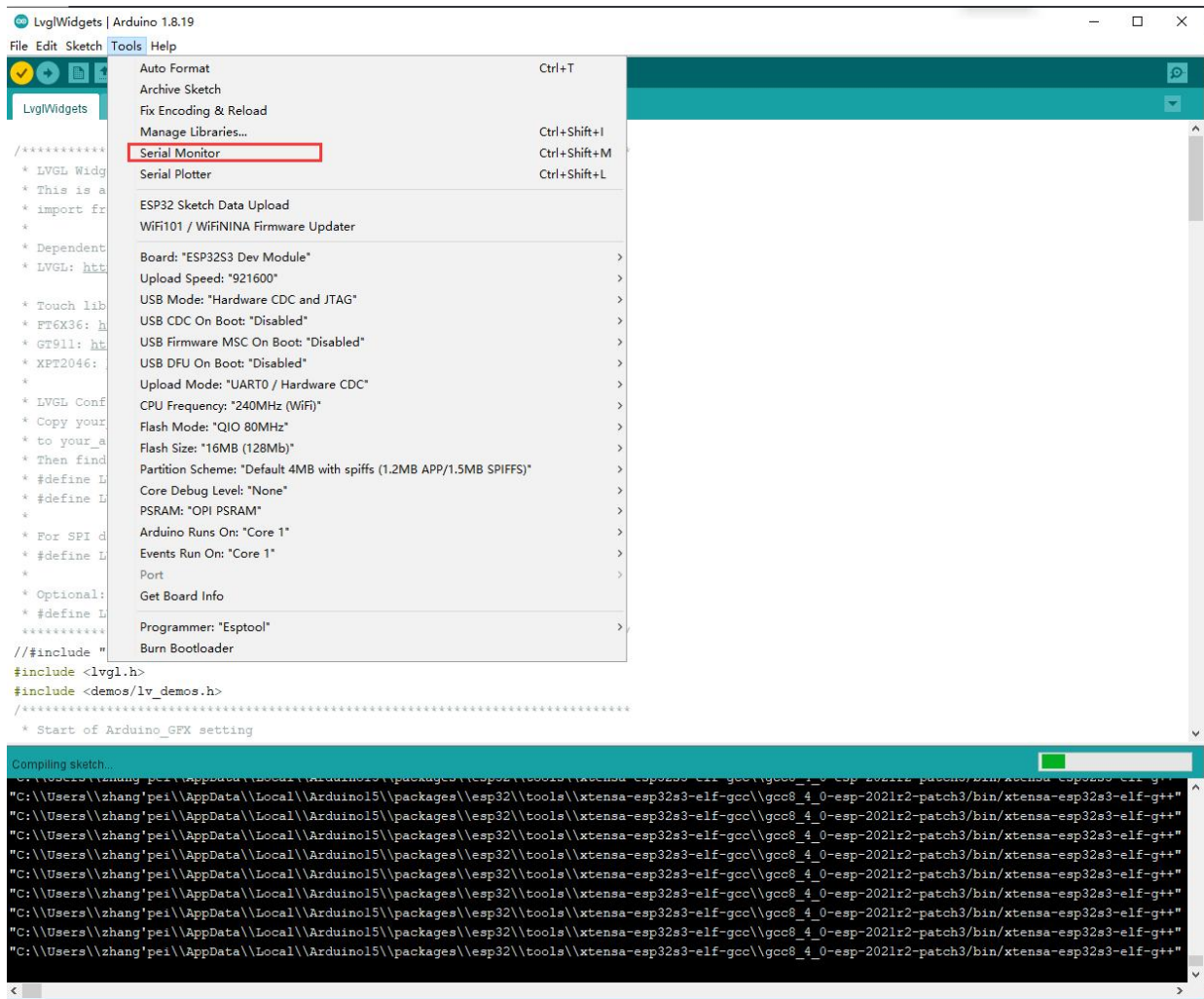
It's important to note that after the code is uploaded, the device will start to run it. So, if we want to upload a new program, we need to reset the power of the device, in order to guarantee that it enters flashing mode again.

First program

Since this platform is based on Arduino, we can use many of the usual functions. As an example for the first program, the code below starts the Serial port and prints "hello from ESP32" every second.

```
void setup() {  
    Serial.begin(115200);  
}  
  
void loop() {  
    Serial.println("hello from ESP32");  
    delay(1000);  
}
```

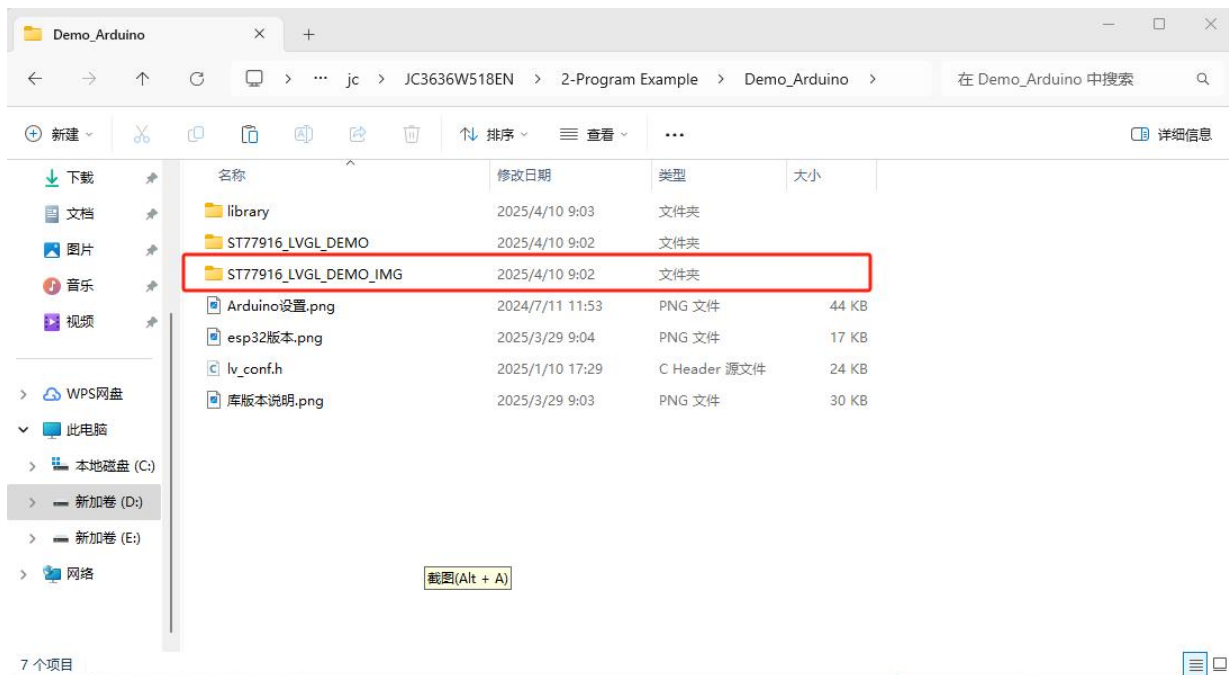
If everything is working fine, we will see the output in the serial console shown.



Again thank you for so much concern.. Hopefully, it's the beginning of a wonderful relationship!

Sample program usage

At present, only a preliminary explanation and introductory use are given to the samples displayed on the screen, and the corresponding examples in the data center are found, as shown in the figure:

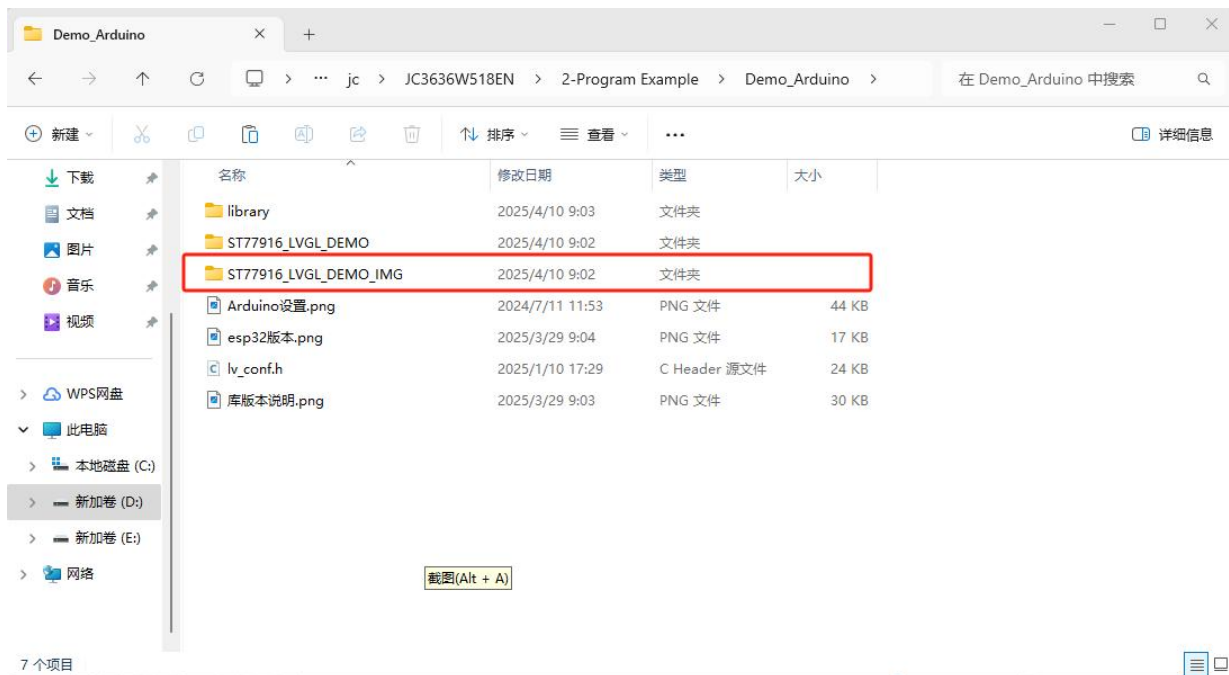


The examples in the red circle are all based on the ESP32_Display_Panel library and ESP32_IO_Expander library and LVGLlibrary.

About the use of touch and LVGL:

Find the data center ST77916_LVGL_DEMO_IMG

As shown:

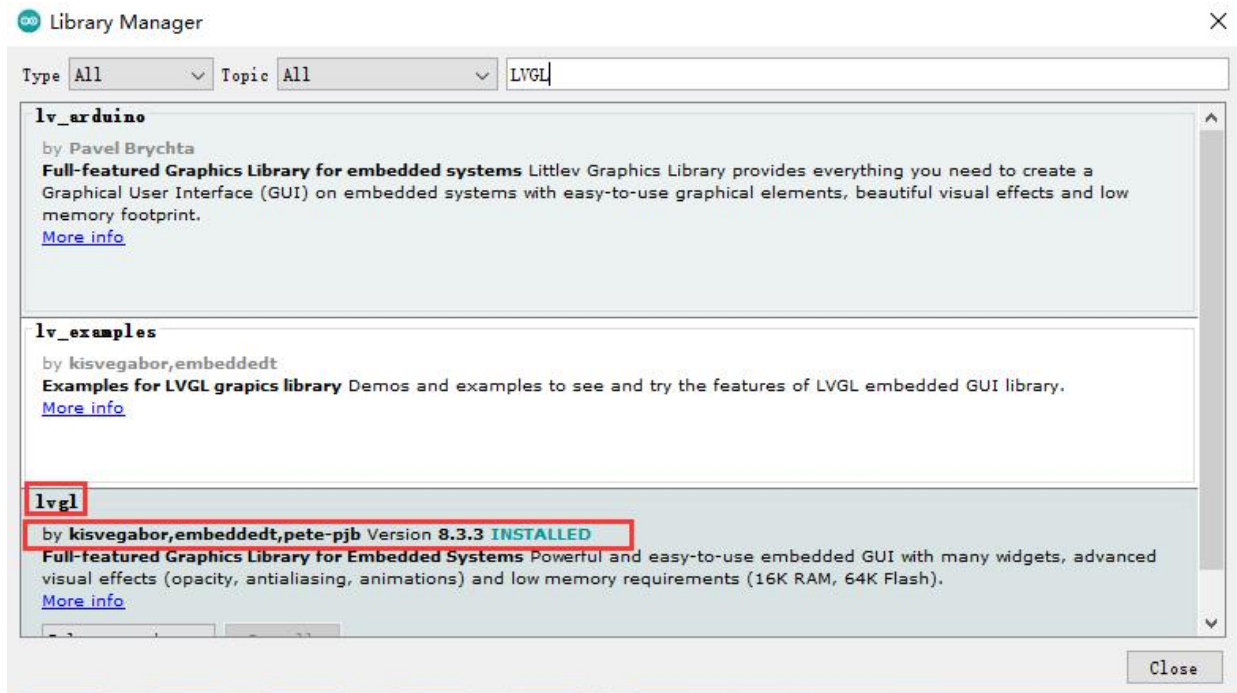


Download three library files .

Requires Arduino 3.0.1 and above



Install the ESP32_Screen (1.0.1) library and
ESP32_IO_Expander(1.1.0)library
LVGL installation library 8.4.0 and below
One -Lvgl



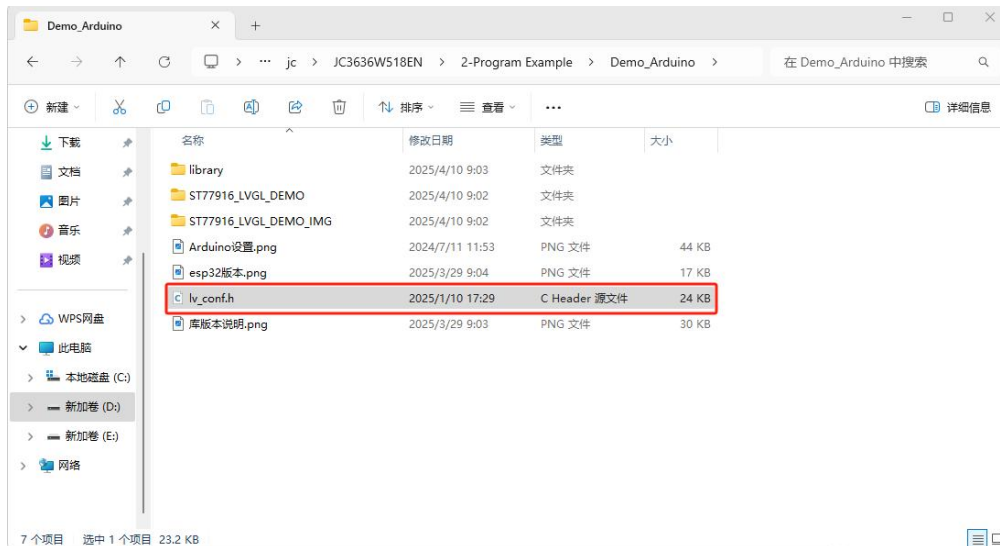
Two -ESP32_Display_Panel and Three -ESP32_IO_Expander





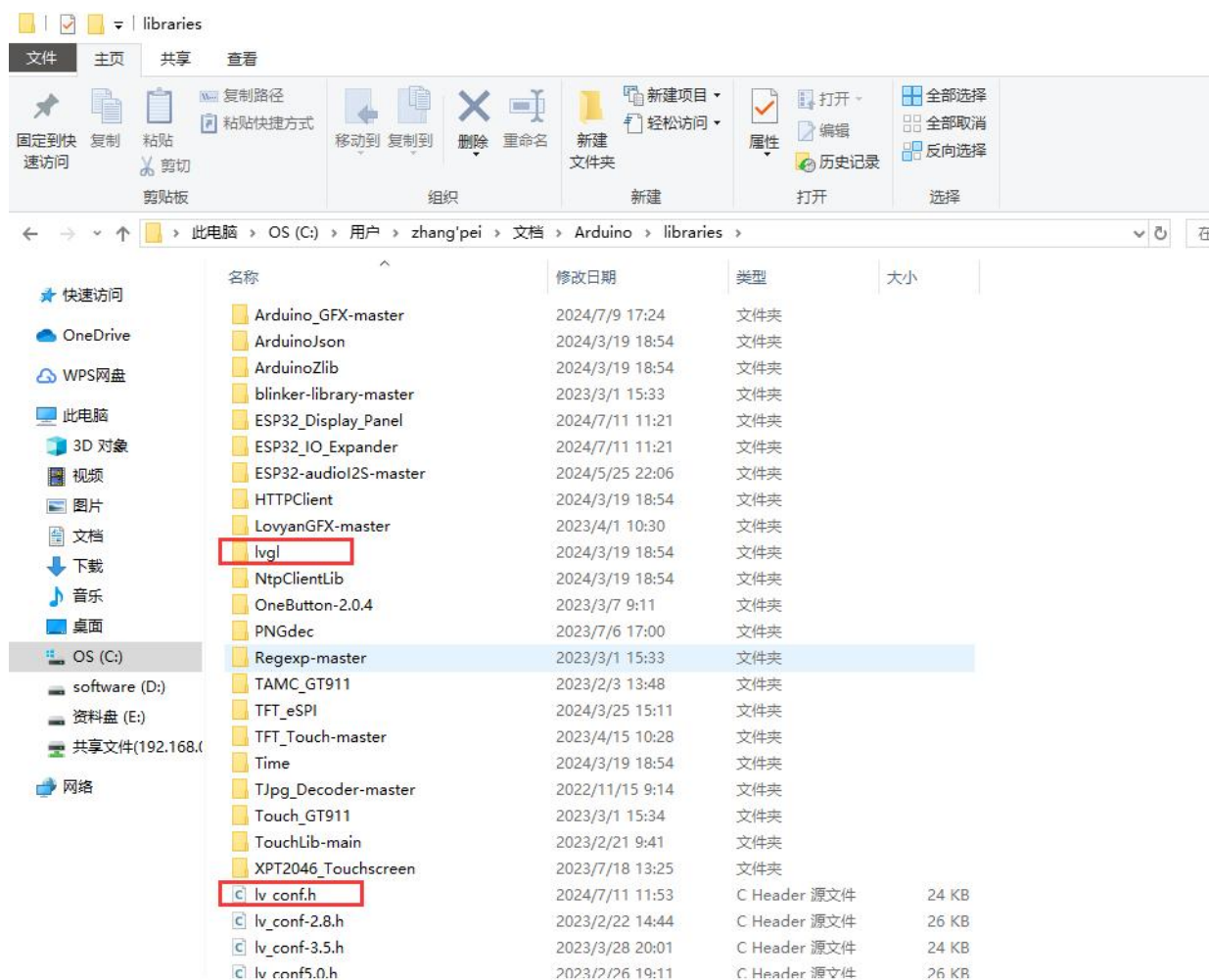
Copy the lv_conf.h of the data center .

As shown:

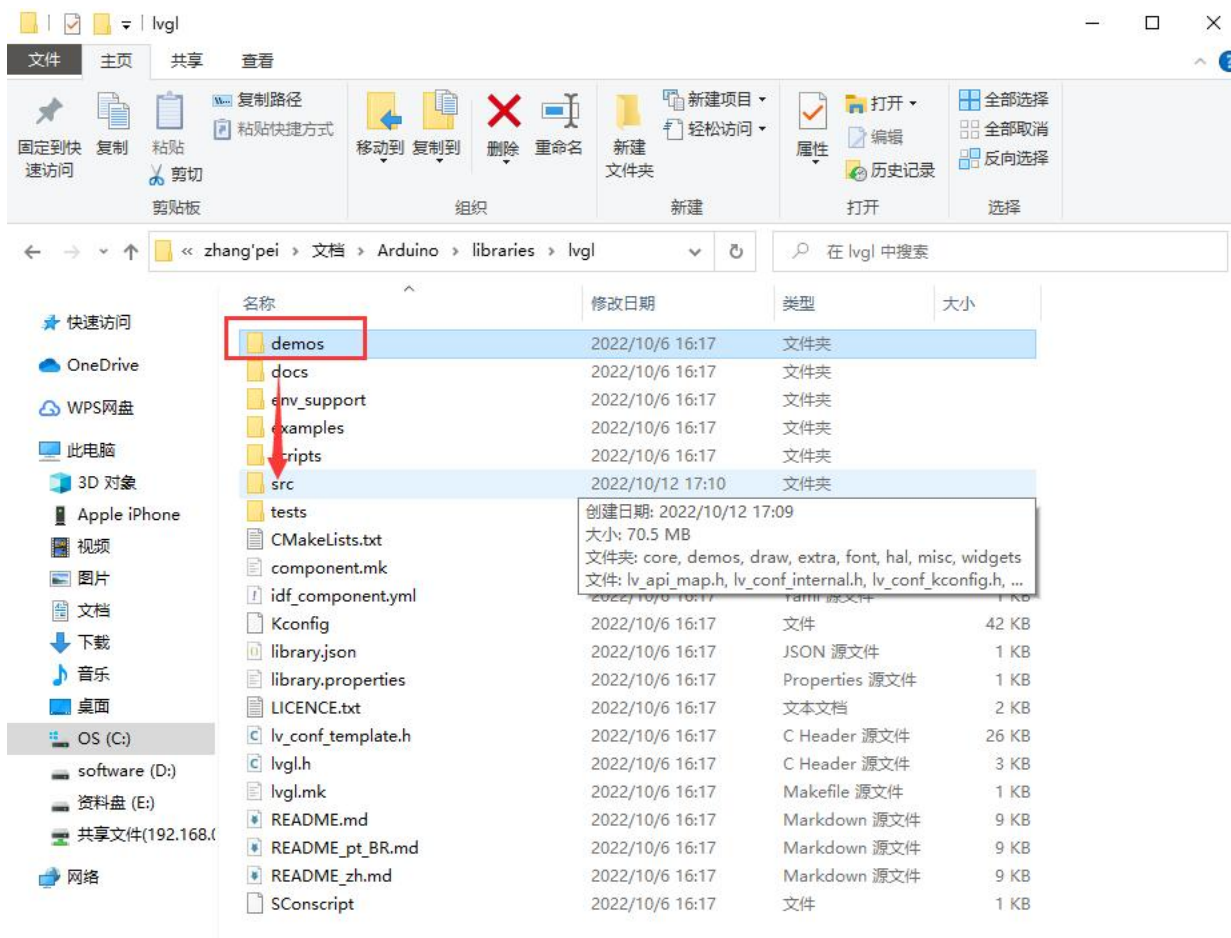


Put this file under the arduino library file, it must be in the same root directory as the library lvgl .

As shown:



Three-Lvgl demos The file is copied to the SRC folder
As shown:



After compiling, you can run LVGL and touch normally.